

Sequences

Real Analysis

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Sequences

Limit Theorems

Monotone
Sequences

Subsequences

The Cauchy
CriterionInterchange of
Limits

Definition

A **sequence of real numbers** (or a **sequence in $\mathbb{R}$**) is a function defined on the set $\mathbb{N} = \{1, 2, \dots\}$ of natural numbers whose range is contained in the set $\mathbb{R}$ of real numbers. If $X : \mathbb{N} \rightarrow \mathbb{R}$ is a sequence, we will usually denote the value of X at n by the symbol x_n rather than using the function notation $X(n)$. The values x_n are also called the **terms** or the **elements** of the sequence. We will denote this sequence by the notations

$$X, \quad (x_n), \quad (x_n : n \in \mathbb{N}).$$

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Definition

A sequence $X = (x_n)$ in $\mathbb{R}$ is said to **converge to** $x \in \mathbb{R}$, or x is said to be a **limit of** (x_n) , if for every $\varepsilon > 0$ there exists a natural number $K(\varepsilon)$ such that for all $n \geq K(\varepsilon)$, the terms x_n satisfy $|x_n - x| < \varepsilon$. If a sequence has a limit, we say that the sequence is **convergent**; if it has no limit, we say that the sequence is **divergent**.

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Theorem

Let $X = (x_n)$ be a sequence of real numbers, and let $x \in \mathbb{R}$. The following statements are equivalent.

- (a) X converges to x .*
- (b) For every $\varepsilon > 0$, there exists a natural number K such that for all $n \geq K$, the terms x_n satisfy $|x_n - x| < \varepsilon$.*
- (c) For every $\varepsilon > 0$, there exists a natural number K such that for all $n \geq K$, the terms x_n satisfy $x - \varepsilon < x_n < x + \varepsilon$.*
- (d) For every ε -neighborhood $V_\varepsilon(x)$ of x , there exists a natural number K such that for all $n \geq K$, the terms x_n belong to $V_\varepsilon(x)$.*

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Definition

If $X = (x_1, x_2, \dots, x_n, \dots)$ is a sequence of real numbers and if m is a given natural number, then the **m -tail of X** is the sequence

$$X_m := (x_{m+n} : n \in \mathbb{N}) = (x_{m+1}, x_{m+2}, \dots).$$

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Theorem

Let $X = (x_n : n \in \mathbb{N})$ be a sequence of real numbers and let $m \in \mathbb{N}$. Then the m -tail $X_m = (x_{m+n} : n \in \mathbb{N})$ of X converges if and only if X converges. In this case, $\lim X_m = \lim X$.

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Theorem

Let (x_n) be a sequence of real numbers and let $x \in \mathbb{R}$. If (a_n) is a sequence of positive real numbers with $\lim(a_n) = 0$ and if for some constant $C > 0$ and some $m \in \mathbb{N}$ we have

$$|x_n - x| \leq C a_n \quad \text{for all } n \geq m,$$

then it follows that $\lim(x_n) = x$.

Definition

A sequence $X = (x_n)$ of real numbers is said to be **bounded** if there exists a real number $M > 0$ such that $|x_n| \leq M$ for all $n \in \mathbb{N}$.

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Theorem

A convergent sequence of real numbers is bounded.

Theorem

Let $X = (x_n)$ be a sequence in $\mathbb{R}$ that converges to x . Then

- a) If $Y = (y_n)$ is a sequence of real numbers that converges to y and if $c \in \mathbb{R}$. Then the sequences $X + Y$, $X - Y$, $X \cdot Y$, and cX converge to $x + y$, $x - y$, xy , and cx , respectively.
- b) If $Z = (z_n)$ is a sequence of nonzero real numbers that converges to z and if $z \neq 0$, then the quotient sequence X/Z converges to x/z .

Theorem

If $X = (x_n)$ is a convergent sequence of real numbers and if $x_n \geq 0$ for all $n \in \mathbb{N}$, then $x = \lim(x_n) \geq 0$.

Theorem

If $X = (x_n)$ and $Y = (y_n)$ are convergent sequences of real numbers and if $x_n \leq y_n$ for all $n \in \mathbb{N}$, then $\lim(x_n) \leq \lim(y_n)$.

Theorem

If $X = (x_n)$ is a convergent sequence and if $a \leq x_n \leq b$ for all $n \in \mathbb{N}$, then $a \leq \lim(x_n) \leq b$.

Theorem

Let the sequence $X = (x_n)$ converge to x . Then the sequence $(|x_n|)$ of absolute values converges to $|x|$. That is, if $x = \lim(x_n)$, then $|x| = \lim(|x_n|)$.

Theorem

Let $X = (x_n)$ be a sequence of real numbers that converges to x and suppose that $x_n \geq 0$. Then the sequence $(\sqrt{x_n})$ of positive square roots converges and $\lim(\sqrt{x_n}) = \sqrt{x}$.

Theorem

Let (x_n) be a sequence of positive real numbers such that $L := \lim(x_{n+1}/x_n)$ exists. If $L < 1$, then (x_n) converges and $\lim(x_n) = 0$.

Definition

Let $X = (x_n)$ be a sequence of real numbers. We say that X is **increasing** if it satisfies the inequalities

$$x_1 \leq x_2 \leq \cdots \leq x_n \leq x_{n+1} \leq \cdots .$$

We say that X is **decreasing** if it satisfies the inequalities

$$x_1 \geq x_2 \geq \cdots \geq x_n \geq x_{n+1} \geq \cdots .$$

We say that X is **monotone** if it is either increasing or decreasing.

Theorem (Monotone Convergence Theorem)

*A monotone sequence of real numbers is convergent if and only if it is bounded.
Further:*

(a) *If $X = (x_n)$ is a bounded increasing sequence, then*

$$\lim(x_n) = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) *If $Y = (y_n)$ is a bounded decreasing sequence, then*

$$\lim(y_n) = \inf\{y_n : n \in \mathbb{N}\}.$$

Definition

Let $X = (x_n)$ be a sequence of real numbers and let $n_1 < n_2 < \cdots < n_k < \cdots$ be a strictly increasing sequence of natural numbers. Then the sequence $X' = (x_{n_k})$ given by

$$\{x_{n_1}, x_{n_2}, \dots, x_{n_k}, \dots\}$$

is called a **subsequence of X** .

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Theorem

If a sequence $X = (x_n)$ of real numbers converges to a real number x , then any subsequence $X' = (x_{n_k})$ of X also converges to x .

Theorem

Let $X = (x_n)$ be a sequence of real numbers. Then the following are equivalent:

- (i) The sequence $X = (x_n)$ does not converge to $x \in \mathbb{R}$.
- (ii) There exists an $\varepsilon_0 > 0$ such that for any $k \in \mathbb{N}$, there exists $n_k \in \mathbb{N}$ such that $n_k \geq k$ and $|x_{n_k} - x| \geq \varepsilon_0$.
- (iii) There exists an $\varepsilon_0 > 0$ and a subsequence $X' = (x_{n_k})$ of X such that $|x_{n_k} - x| \geq \varepsilon_0$ for all $k \in \mathbb{N}$.

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Theorem (The Bolzano-Weierstrass Theorem)

A bounded sequence of real numbers has a convergent subsequence.

Theorem

Let $X = (x_n)$ be a bounded sequence of real numbers and let $x \in \mathbb{R}$ have the property that every convergent subsequence of X converges to x . Then the sequence X converges to x .

Definition

Let $X = (x_n)$ be a bounded sequence of real numbers.

- (a) The **limit superior of** (x_n) is the infimum of the set V of $v \in \mathbb{R}$ such that $v < x_n$ for at most a finite number of $n \in \mathbb{N}$. It is denoted by

$$\limsup(x_n) \quad \text{or} \quad \limsup X \quad \text{or} \quad \overline{\lim}(x_n).$$

- (b) The **limit inferior of** (x_n) is the supremum of the set of $w \in \mathbb{R}$ such that $x_m < w$ for at most a finite number of $m \in \mathbb{N}$. It is denoted by

$$\liminf(x_n) \quad \text{or} \quad \liminf X \quad \text{or} \quad \underline{\lim}(x_n).$$

Theorem

If (x_n) is a bounded sequence of real numbers, then the following statements for a real number x^ are equivalent.*

- (a) $x^* = \limsup(x_n)$.
- (b) *If $\varepsilon > 0$, there are at most a finite number of $n \in \mathbb{N}$ such that $x^* + \varepsilon < x_n$, but an infinite number of $n \in \mathbb{N}$ such that $x^* - \varepsilon < x_n$.*
- (c) *If $u_m = \sup\{x_n : n \geq m\}$, then $x^* = \inf\{u_m : m \in \mathbb{N}\} = \lim(u_m)$.*
- (d) *If S is the set of subsequential limits of (x_n) , then $x^* = \sup S$.*

Theorem

A bounded sequence (x_n) is convergent if and only if $\limsup(x_n) = \liminf(x_n)$.

Definition

A sequence $X = (x_n)$ of real numbers is said to be a **Cauchy sequence** if for every $\varepsilon > 0$ there exists a natural number $H(\varepsilon)$ such that for all natural numbers $n, m \geq H(\varepsilon)$, the terms x_n, x_m satisfy $|x_n - x_m| < \varepsilon$.

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Lemma

If $X = (x_n)$ is a convergent sequence of real numbers, then X is a Cauchy sequence.

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Lemma

A Cauchy sequence of real numbers is bounded.

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Theorem (Cauchy Convergence Criterion)

A sequence of real numbers is convergent if and only if it is a Cauchy sequence.

Definition

We say that a sequence $X = (x_n)$ of real numbers is **contractive** if there exists a constant C , $0 < C < 1$, such that

$$|x_{n+2} - x_{n+1}| \leq C|x_{n+1} - x_n|$$

for all $n \in \mathbb{N}$. The number C is called the **constant of the contractive sequence**.

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Theorem

Every contractive sequence is a Cauchy sequence, and therefore is convergent.

Corollary

If $X := (x_n)$ is a contractive sequence with constant C , $0 < C < 1$, and if $x^* := \lim X$, then

$$(i) \quad |x^* - x_n| \leq \frac{C^{n-1}}{1-C} |x_2 - x_1|,$$

$$(ii) \quad |x^* - x_n| \leq \frac{C}{1-C} |x_n - x_{n-1}|.$$

Definition

Let (x_n) be a sequence of real numbers.

- (i) We say that (x_n) **tends to** $+\infty$, and write $\lim(x_n) = +\infty$, if for every $a \in \mathbb{R}$ there exists a natural number $K(a)$ such that if $n \geq K(a)$, then $x_n > a$.
- (ii) We say that (x_n) **tends to** $-\infty$, and write $\lim(x_n) = -\infty$, if for every $b \in \mathbb{R}$ there exists a natural number $K(b)$ such that if $n \geq K(b)$, then $x_n < b$.

We say that (x_n) is **properly divergent** in case we have either $\lim(x_n) = +\infty$ or $\lim(x_n) = -\infty$.

Theorem

A monotone sequence of real numbers is properly divergent if and only if it is unbounded.

- (a) *If (x_n) is an unbounded increasing sequence, then $\lim(x_n) = +\infty$.*
- (b) *If (x_n) is an unbounded decreasing sequence, then $\lim(x_n) = -\infty$.*

Theorem

Let (x_n) and (y_n) be two sequences of real numbers and suppose that

$$x_n \leq y_n \quad \text{for all } n \in \mathbb{N}.$$

(a) *If $\lim(x_n) = +\infty$, then $\lim(y_n) = +\infty$.*

(b) *If $\lim(y_n) = -\infty$, then $\lim(x_n) = -\infty$.*

Theorem

Let (x_n) and (y_n) be two sequences of positive real numbers and suppose that for some $L \in \mathbb{R}$, $L > 0$, we have

$$\lim(x_n/y_n) = L.$$

Then $\lim(x_n) = +\infty$ if and only if $\lim(y_n) = +\infty$.

Definition

Let $A \subseteq \mathbb{R}$ and suppose that for every $n \in \mathbb{N}$ there exists a function $f_n : A \rightarrow \mathbb{R}$; we say that (f_n) is a **sequence of functions from A to $\mathbb{R}$** .

Definition

Let (f_n) be a sequence of functions from $A \subseteq \mathbb{R}$ to $\mathbb{R}$, let $A_0 \subseteq A$, and let $f : A_0 \rightarrow \mathbb{R}$. We say that **the sequence (f_n) converges to f on A_0** if, for every $x \in A_0$, the sequence $(f_n(x))$ converges to $f(x)$ in $\mathbb{R}$. In this case f is called the **limit of the sequence (f_n) on A_0** . When such a function exists, we say that the sequence (f_n) is **convergent on A_0** , or that (f_n) **converges pointwise on A_0** .

Remark

To denote that the sequence (f_n) converges to f on A_0 , we sometimes write

$$f = \lim(f_n) \text{ on } A_0, \quad \text{or} \quad f_n \rightarrow f \text{ on } A_0.$$

Occasionally, when f_n and f are given by formulas, we write

$$f(x) = \lim f_n(x) \text{ for } x \in A_0, \quad \text{or} \quad f_n(x) \rightarrow f(x) \text{ for } x \in A_0.$$

Lemma

A sequence (f_n) of functions from $A \subseteq \mathbb{R}$ to $\mathbb{R}$ converges to a function $f : A_0 \rightarrow \mathbb{R}$ if and only if for every $\varepsilon > 0$ and for every $x \in A_0$ there exists a natural number $K(\varepsilon, x)$ such that if $n \geq K(\varepsilon, x)$, then

$$|f_n(x) - f(x)| < \varepsilon.$$

Definition

A sequence (f_n) of functions from $A \subseteq \mathbb{R}$ to $\mathbb{R}$ **converges uniformly on** $A_0 \subseteq A$ **to a function** $f : A_0 \rightarrow \mathbb{R}$ if for every $\varepsilon > 0$ there exists a natural number $K(\varepsilon)$ such that if $n \geq K(\varepsilon)$, then

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in A_0.$$

In this case we say that the sequence (f_n) is **uniformly convergent on** A_0 . We sometimes write

$$f_n \Rightarrow f \text{ on } A_0, \quad \text{or} \quad f_n(x) \Rightarrow f(x) \text{ for } x \in A_0.$$

Lemma

A sequence (f_n) of functions from $A \subseteq \mathbb{R}$ to $\mathbb{R}$ does not converge uniformly on $A_0 \subseteq A$ to a function $f : A_0 \rightarrow \mathbb{R}$ if and only if for some $\varepsilon_0 > 0$ there exists a subsequence (f_{n_k}) of (f_n) and a sequence (x_k) in A_0 such that

$$|f_{n_k}(x_k) - f(x_k)| \geq \varepsilon_0 \quad \text{for all } k \in \mathbb{N}.$$

Definition

If $A \subseteq \mathbb{R}$ and $\varphi : A \rightarrow \mathbb{R}$ is a function, we say that φ is **bounded on A** if the set $\varphi(A)$ is a bounded subset of $\mathbb{R}$. If φ is bounded, we define the **uniform norm of φ on A** by

$$\|\varphi\|_A := \sup\{|\varphi(x)| : x \in A\}.$$

Lemma

A sequence (f_n) of bounded functions on $A \subseteq \mathbb{R}$ converges uniformly to f on A if and only if $\|f_n - f\|_A \rightarrow 0$.

Theorem (Cauchy Criterion for Uniform Convergence)

Let (f_n) be a sequence of bounded functions on $A \subseteq \mathbb{R}$. Then this sequence converges uniformly to a bounded function f on A if and only if for every $\varepsilon > 0$ there exists a number $H(\varepsilon)$ in $\mathbb{N}$ such that for all $m, n \geq H(\varepsilon)$, then $\|f_m - f_n\|_A \leq \varepsilon$.

Theorem

Let (f_n) be a sequence of continuous functions on a set $A \subseteq \mathbb{R}$ and suppose that (f_n) converges uniformly to a function $f : A \rightarrow \mathbb{R}$ on A . Then f is continuous on A .

Theorem

Let $J \subseteq \mathbb{R}$ be a bounded interval and let (f_n) be a sequence of functions from J to $\mathbb{R}$. Suppose that there exists $x_0 \in J$ such that $(f_n(x_0))$ converges and that the sequence (f'_n) of derivatives exists on J and converges uniformly to a function g on J . Then the sequence (f_n) converges uniformly to a function f on J that has a derivative at every point of J and $f' = g$.

Theorem

Let (f_n) be a sequence of functions in $\mathcal{R}[a, b]$ and suppose that (f_n) converges uniformly to f on $[a, b]$. Then $f \in \mathcal{R}[a, b]$ and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

Theorem (Bounded Convergence Theorem)

Let (f_n) be a sequence of functions in $\mathcal{R}[a, b]$ that converges on $[a, b]$ to a function $f \in \mathcal{R}[a, b]$. Suppose also that there exists $B > 0$ such that $|f_n(x)| \leq B$ for all $x \in [a, b]$, $n \in \mathbb{N}$. Then

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

Theorem (Dini's Theorem)

Suppose that (f_n) is a monotone sequence of continuous functions on $I := [a, b]$ that converges on I to a continuous function f . Then the convergence of the sequence is uniform.